Table des matières

TRAUPIXE Manual V25.10

I. Introduction :

TrauPIXE enables to process the multiple PIXE spectra combination and numerous spectra extracted from map or saved during a full day of analyses.
The general principle of TrauPIXE is to use the Gupixwin process engine , executable “pixwin.exe”, to process each spectrum with different processing solution (matrix or trace). To do that TrauPIXE interact dynamically with the input/output file used by the executable, the default spectrum and parameters are edited into the “gupiwin.par” then output files such as “pixstats.out” or “pixoxide.out” are read to extract results.

II. Installation

TrauPIXE is developed using Visual Basic (V 2022)and utilizing sub-DLLs created with LabVIEW National Instruments, requires no installation process.
Simply copy the folder containing all DLLs and the TrauPIXE.exe software.
Create a folder c:\tmp_TrauPIXE and give all right access on it.

A licenced version of Gupixwin (2.23 or 3.0) is mandatory.

Two mandatory runtimes are necessary:

1 - ".NET 6.0.XX Desktop Runtime"
Download here : https://dotnet.microsoft.com/en-us/download/dotnet/thank-you/runtime-desktop-6.0.27-windows-x86-installer

2 - "LabviewRunTime 2016 32Bit" to save Fit data as PNG File.(Bug if not present)
Download here : https://www.ni.com/fr/support/downloads/software-products/download.labview-runtime.html#310816

III. Spectrum file format accepted by TrauPIXE

One file format is taken in charge by TrauPIXE : The GUPIXWIN Ascii File Structure A single header line contains first the number of channel and second a zero. All subsequent lines

contain the counts for each channel; an example follows, subject to rules given below. The zero in the first line indicates that there are no lines of text present.

First format:

2048 0
15
10
32
21
etc.

The following are acceptable delimiters between channel contents and also between entries within the header: space, tab, decimal point/period, comma. Any number of approved delimiters may appear between values. Channel content values can occur any number to a line. First header values (e.g. 1024 0) must be on a single line. GUPIXWIN cannot accommodate real number channel contents such as 215.71, because such an entry would be interpreted as two readings, namely 215 with a period delimiter followed by 71; the inclusion of real numbers will generate an error message signalling that too many channels were found.

Second format :

Has an extra line of information between the first line and the channel count lines. shows year, day of year,time of day (in seconds past midnight), measurement duration in seconds, total spectral counts, and a file ID of up to 31 characters enclosed in quotes
Example:

2048 1
2024 02 65312 122 181674 'Mesure de charge ,1000,1000,25,25,500,40,50000,Proton , 2995 keV , 40mm He , 50 um Al , OFF , 50 um Al , 100 um Al'
15
10
32
21
etc.
20
etc.

IV. « config-TrauPIXE.ini » file

This file is and must be located into the TRAUPIXE.EXE folder to be loaded.
It contained the file format or file extension of any facilities.
It also defined the path for Gupix folder [GUPIX PATH] tag.
Once loaded it custom the TrauPIXE by defined the file format/extension of spectrum files.
It will first try to find spectrum with corresponding extension, if it doesn't find any file it will try to find filename with "tag" in the name.

In the following example :
For [DET0] , It will try to find "*.x0" or "*X0*.*"
Correct for AGLAE spectra named "20240102_001_prj_obj.x0 , 20240102_002_prj_obj.x0, ….

For DET1 , It will try to find "*.romas" or "*ROMAS*.*"
Correct with spectra named "ROMAS001.DAT, ROMAS002.DAT, ……

Example of a "Config_TrauPIXE.ini"

```
[GUPIX PATH]
c:\gupixwin\gupix

[DET0]
X0
[DET1]
ROMAS
[DET2]
X10
[DET3]
X13
[DET4]
X4
[DET5]
X3
[DET6]
X12
[DET7]
X11
[DET8]
X2
```

V. « external-conc.csv » file

Only used for “S_xxxxx” output sheet.

This file is used to consider automatically external concentrations during the PIXE processing.
Indeed, elemental concentrations obtained by PIGE, RBS or others techniques can be given as an input through an external file named ‘external-conc.csv’.

The PIXE concentration will be replaced if a concentration is given for the current analyses and present in PIXE (i.e., Na_2O) or will be added to the matrix composition if they are not detected by PIXE (H_2O, B_2O_3, F, CO_2, Li_2O, BeO). The new matrix composition is used to define the matrix for the trace processing, invisible element are also used to loop concentration to 100%

If no value is given or 0 value then the PIXE concentration will be kept.

example:
Z;11;3;6;12;4
Formula;Na2O;Li;B;MgO;CO2
Type;PIGE;PIGE;EXT;STD;STD
20220825_0003_std_VERRE_IBA;30000;0;0;0;
20220825_0005_std_VERRE_IBA;0;4903;500000;1452;
20220825_0006_std_VERRE_IBA;75000;;;0;7500

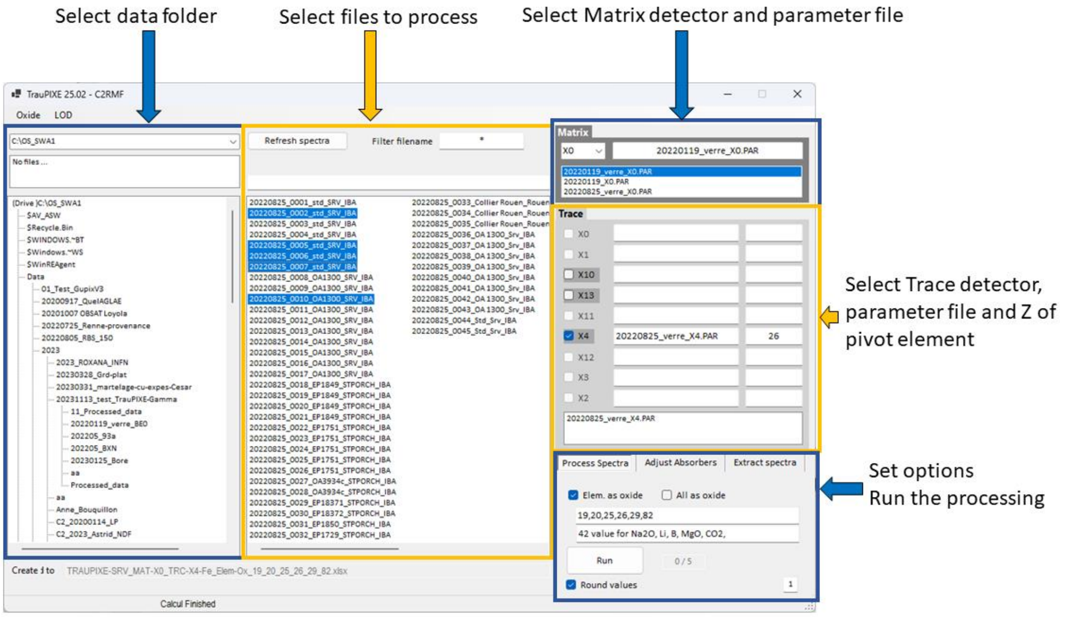
VI. « charge-exp.csv» file

Use external charge value instead of Pivot method.

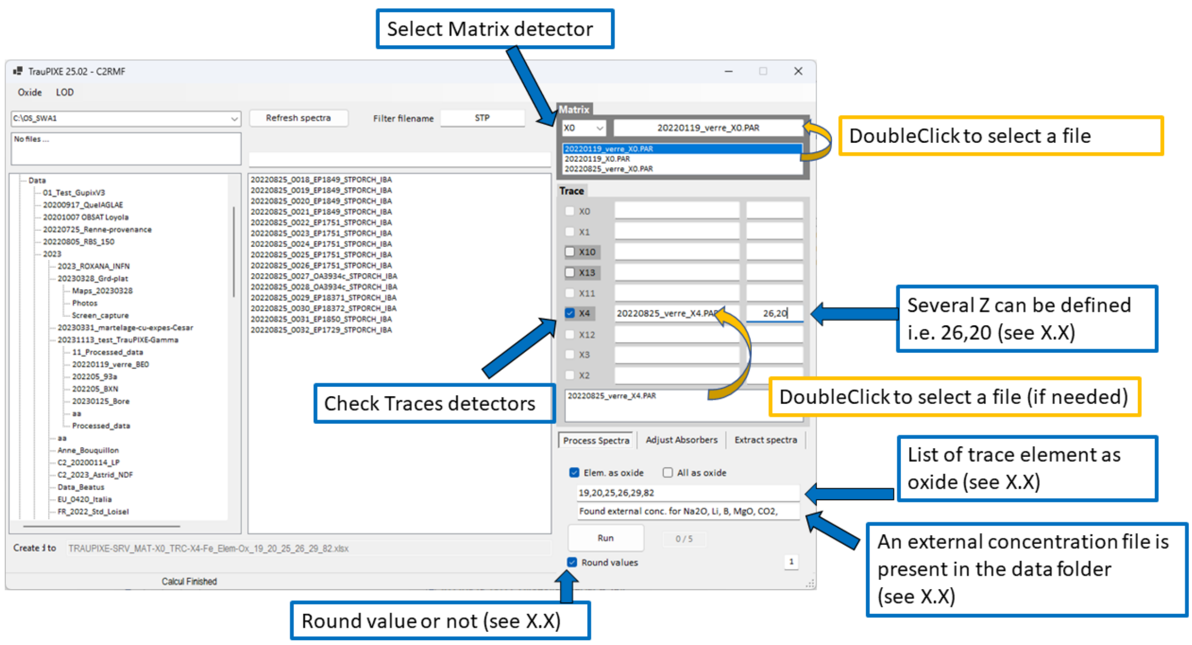
example:
FILENAME;Q ROMAB;Q ROMAS
ROMA002;0,0366;0,0371
ROMA004;0,0319;0,0335
ROMA005;0,1709;0,1674
ROMA007;0,187;0,1823

VII. Interface

1- General overview of TrauPIXE user interface



2- Details on TrauPIXE user interface



Notes: When you select the Matrix detector or check a trace detector : if only one parameter file meets file naming requirements, it is automatically selected.

VIII. Methodology to process Matrix and Trace spectra with the Pivot method

1- Pivot method

The physics engine of GUPIX software is used to process the PIXE spectra obtained with the different detectors. The processing of the spectra requires an accurate knowledge of the experimental setup (i.e., cumulated charge on the sample during the irradiation, solid angle of the detectors, absorbers placed in front of the detectors, dead time, etc.). The difficulties with an external beam setup are that the integrated charge cannot be accurately measured and that solid angle of the detectors is often modified for each run to optimize the count-rate ratio between the trace signal and the matrix signal, depending on the type of matrix composition and the absorbers. The limitation of undetermined parameters (cumulated charge and solid angle) is not a problem for the matrix processing in the case of CH samples since the "Iterated Matrix Solution" of GUPIX can be used. This process normalizes the sum of the concentrations of the matrix elements to 100 % as we assume that we detect all matrix elements in a thick sample. For the trace processing, a different approach is required to solve this issue, which is why we use the fixed matrix solution of GUPIX. First, for each trace spectrum obtained from the different trace detectors, the matrix composition must be defined with the one obtained during the matrix processing and secondly, we must find a way to calculate the normalization concentration factor. For that, we use the pivot method already mentioned in previous works. This method consists in processing the spectra of the traces with an arbitrary charge and solid angle. Thus, we obtain arbitrary concentrations of the traces in the material, but with the correct concentration ratio. Therefore, if the concentration of one trace element is known, it can be used to scale all the other concentrations. This particular element must be present with high statistics in the matrix spectrum and in all the trace spectra, and it is known as the pivot element. We have to point out that the uncertainty of this element is critical as it will be propagated to all the other trace elements. The composition of the matrix and the X-ray transmission of the absorbers installed on the detectors are also important factors to consider when selecting the pivot element, for example, usually lead is chosen in the case of leaded glass, or copper in the case of bronze. In some cases, such as metallic objects, one can obtain different matrix composition that can change from one point to another. It may be essential to automatically select the appropriate pivot elements from a predefined list, which is a function included in TrauPIXE.

Once the concentration of the pivot element is obtained from both the matrix and trace spectra, correction factor for the cumulated charge is obtained following

2- Equation of the correction factor to normalize trace:

$$Q' = Q \cdot \frac{PivotConcTrace}{PivotConcMatrix} \tag{1}$$

Where *Q'* is the charge factor that will be used in GUPIX for the current trace spectrum, *Q* is the charge factor used in the first trace processing (in our case it is arbitrary since we do not know the real cumulated charge), *PivotConcMatrix* is the concentration of the pivot element obtained from the matrix spectrum, and *PivotConcTrace* is the concentration of the pivot element obtained from the trace spectra.

After replacing the new charge factor in the trace parameter file, a second trace processing is run, providing normalized trace concentrations.

IX. Details on TrauPIXE features

1- Round of the concentration value

Concentration value are rounded at the hundreds of ppm for value above 10% or 100000ppm, tens of ppm between 1000 and 99999 ppm and remain unrounded for low concentrations (<1000 ppm).

example :
"154251" round as "154300" (max difference 50 ppm for 100000ppm, 0.05%)
"5415" round as "5410" (max difference 5 ppm for 1000ppm, 0.5%)
"584" is not rounded

2- A smart selection in case of several Pivot element

For each trace detector, multiple Z pivots can be set (e.g., "20,26,82"). A function selects, for each analysis, the one with the lowest uncertainty by comparing the Fit error given by GUPIX.

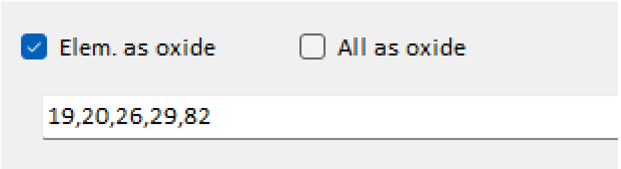
If its uncertainty (quadratic sum of Matrix and Trace fit error) is the lowest then the element is selected as Pivot.
It means it can change for every point and can be different for each trace detector. For example, to normalize trace concentration for trace X4 it can selects 26 (Fe) and for trace X13 it can selects 82 (Pb).

Update V25.05 (May 2024): The Column "Z Pivot" in the "Exp. Data" sheet indicate witch element have been selected to do Pivot.

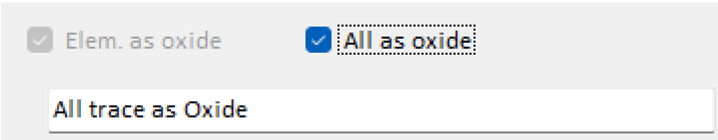
Before V25.05: For the moment any indicator is given to see witch element have been used. A simple comparation between the value in matrix and trace give this information as their concentration are equal.

3- What is the Text “Elem. as oxide” ?

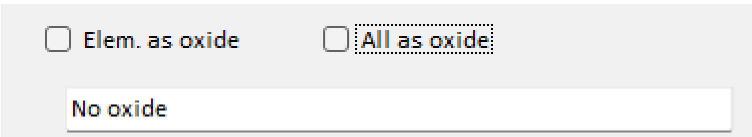
In this text box the user can set element as oxide as a list of atomic number :



Or All as oxide :



Or All in elemental / No oxide:



This list of atomic number concern only elements present in both spectra matrix and trace.
If you want to set matrix element in elemental form you must edit them inside the ASCII parameter file (.par) or with Gupixwin software.
Note that the Matrix element form (elemental, oxide or other complex compound) are defined into the Matrix parameter file. (If O is defined as invisible element they are by default in oxide form.)

In the final result table (see paragraph 6) the table could mixt value in oxide and some others trace in elemental form.

MgO	Al2O3	SiO2	SO3	K2O	CaO	TiO2	Mn	Fe2O3	As2O3
61000	135800	613700	13740	42110	46560	20470	381	256	11740

Here Mn is present in matrix and trace parameter file and have not been requested as oxide.

4- Selection of the best concentration in “S_” worksheet

After the matrix and trace processing, TrauPIXE select only one concentration for each element.
Do to it, it compares for each element their respective uncertainty.

For matrix element their uncertainty is the direct value fit error provide by GUPIX as for trace element their uncertainty or Trace-total-error added the Total-pivot-error.
Where:
Total-pivot-error = Sqrt (Matrix-pivot-fit-error^2 + Trace-pivot-fit-error^2)
Matrix-pivot-fit-error = Gupix Fit error of the Matrix pivot element.
Trace-pivot-fit-error = Gupix Fit error of the Trace pivot element.

example:
If you select Fe as Pivot element. If Fe have a fit error of 2.5% in the matrix spectra and a Fit error or 8% in the trace spectra the Total-pivot-error is 8.38%.
This Pivot uncertainty are added for each trace element to their direct Fit error as :
Trace-element-error = Sqrt (Total-pivot-error^2 + Element-fit-error^2)
example :
If a trace element have a low fit-error (1.5%) due to the high uncertainty of the pivot element (8.38%) its total error will be 8.51%.

It clearly illustrate that the selection of a “good” pivot element is critical to reduce as much as possible the uncertainty on traces results.

5- An exception in best concentration selection

Due to some incorrect GUPIX configuration some unrealistic concentration or LOD could be calculated by GUPIX.
For example, at AGLAE, it often appends in the case of asking PbM concentration into a trace detector where an absorber of 50µm Al or 100 µm Al is installed. It results a detectable element (not tagged as “N”) with a very high LOD. Those in all “**S_”** sheets this false and very high value (10% or more) induce a wrong result for all others concentrations as its concentration is consider during the 100% normalization.

To avoid this TrauPIXE ignore element tagged as “?” showing a concentration > 50000 ppm or 5 % (before V25.08 ; oct. 2024 : 9999 ppm) . It seems logical that if an element is present at this level of concentration it should be well measured so tagged as “Y”.

6- Classic output sheet and new sheets “S_” (2023)

GUPIX output results (last table in “pixstat.out” and “pixoxide.out”) are provided in several excel sheets:
Respectively named:
Elemental Conc. ; Oxide Conc., LOD, Peak Height, Area, Fit-error.
Additionally to this “classic” output TrauPIXE provide new output results

corresponding to the new function of smart selection of the best value (See IV.4):

a) “S_Conc ppm”
Concentration in ppm unit
Concentration tagged as “?” are considered correct despite their concentration are <LOQ (with a few exceptions see chapter V) ,
Value “N” are not reported instead it provide the “<LOD”

b) “S_Conc %”
Same than a) but concentration are expressed in %

c) “S_Conc. & Err. ppm”
Same than a) but next each elemental concentration is reported the % of uncertainty.
Uncertainty is raw Gupix Fit error for matrix or Trace-element-error for traces elements.

d) “S_Conc. & Err. %”
Same than c) but concentration are expressed in %

e) “S_Conc. ppm (RED)”
Same than a) but concentration tagged as “?” are reported as “<LOQ” or “<3.3*LOD”.

f) “S_Conc. % (RED)”
Same than f) but concentration are expressed in %.

g) “Total error”
Output of all uncertainty for all detectors.

h) “S_Best Det.”
Indicate witch detector have been selected for “S_” sheets.

<u>Example of some new “S_” sheets:</u>

Na2O	MgO	Al2O3	SiO2	P2O5	SO3	Cl	K2O	CaO	Ti	V
29710	46120	165200	512200	2520	24910	495	17290	70310	6940	210
25140	40710	172400	537300	2180	12590	242	18270	72290	6710	232
45280	1620	178000	694800	18610	2760	372	36280	7350	60	24

“**S_Conc. ppm**” sheet (a)

Na2O	MgO	Al2O3	SiO2	P2O5	SO3	Cl	K2O	CaO	Ti	V
2,97	4,61	16,5	51,2	0,252	2,49	0,049	1,73	7,03	0,69	0,021
2,51	4,07	17,2	53,7	0,218	1,26	0,024	1,83	7,23	0,67	0,023
4,53	0,162	17,8	69,5	1,86	0,28	0,037	3,63	0,73	0,006	0,002

“**S_Conc. %**” sheet (a)

Na2O	MgO	Al2O3	SiO2	P2O5	SO3	Cl	K2O	CaO	Ti	V
29980	46540	166700	516900	<1033	25140	<488	17450	70960	7000	<436
25150	40730	172500	537500	<1040	12600	<554	18280	72320	6710	<416
45470	<1135	178800	697600	18680	2780	<508	36430	7380	<251	<119

“**S_Conc. ppm (RED)**” sheet (e)

Na2O	*Err.%*	**MgO**	*Err.%*	**Al2O3**	*Err.%*	**SiO2**	*Err.%*	**P2O5**	*Err.%*	**SO3**	*Err.%*
29710	*4,02*	46120	*2,54*	165200	*1,28*	512200	*0,75*	2520	*21,58*	24910	*2,98*
25140	*4,39*	40710	*2,69*	172400	*1,24*	537300	*0,75*	2180	*24,79*	12590	*4,23*
45280	*2,88*	1620	*25,66*	178000	*1,11*	694800	*0,71*	18610	*4,97*	2760	*14,15*

“**S_Conc. & Err. ppm**” sheet (c)

Na	Mg	Al	Si	P	S	Cl	K	Ca	Ti
4,02	2,54	1,28	0,75	**21,58**	2,98	**21,66**	1,88	1,04	2,19
4,39	2,69	1,24	0,75	**24,79**	4,23	**41,9**	1,78	1,02	2,43
2,88	**25,66**	1,11	0,71	4,97	14,15	**26,69**	1,42	3,65	**107,93**

“**Total error**” sheet (g)

7- Experimental data Sheet

The “Exp. Data” sheet show some experimental information.

➢ Mat * by : Normalization factor of the matrix concentration.

Example of GUPIX output where Mat * by = 3.11E-04:
In the matrix iteration section it was found that a correction needed to be applied to the H or uC value in order for the concentrations to sum to unity. The concentrations listed in the table above include this correction value. With the user supplied H & uC values alone the conc would be the above values multiplied by 3.111E-04 *(See the GUPIXWIN documentation for further discussion.)*
If the normalization of concentrations to 100% total is not required this value is 1.

➢ Q (trace detector name) = Normalization factor for trace concentration calculated with the pivot method.
➢ Av.(nA) (detector name) = Average current
➢ Chi2 (detector name) = Chi^2 of the spectrum
➢ Res. (detector name) = Energy resolution (eV)
➢ Cnt/sec. (detector name) = Count per second
➢ Z Pivot (detector name) = Z of the pivot element selected (pivot method)

- <u>Filters (detector name)</u> = Absorbers read from the Gupix parameter file (Z, thickness)
- <u>Par file (detector name)</u> = Gupix parameter file used to process spectra

X. Considering PIGE/RBS concentration during PIXE processing (Update May 2024)

A TrauPIXE major update is the integration of element concentrations measured by other companion techniques such as RBS or PIGE during the PIXE processing. This can be a very interesting option for elements such as B, F, Na, Be, Li, C, N that are invisible or hardly detectable for PIXE (for example, in the case of Na, it is detected by PIXE but the quantification is not correct).

To consider PIGE or RBS concentration, it is necessary to incorporate those concentrations into the PIXE resulting tables, replacing the PIXE concentrations or adding new lines for the elements that are not correctly measured or non-detectable with this technique, the result is then normalized to 100 %. This process has been improved by considering automatically those concentrations during the PIXE processing. Indeed, elemental concentrations obtained by PIGE, RBS or others techniques can be given as an input through an external csv file named 'external-conc.csv'. Their concentration will be replaced if they are present in PIXE with incorrect values (i.e., Na_2O) or will be added to the matrix composition if they are not detected by PIXE (H_2O, B_2O_3, F, CO_2, Li_2O, BeO).

Since version 25.08 :

```
All chemical elements can now be defined using external conc.
The software get its Z and load its atomic mass via atomic_mass.csv
The Oxide valence is extracted from the formula fill in its Formula
cell. This valence is used to consider the oxide conc. part that must
be added to the final O concentration of the matrix.

oxide_conc_to_add = initial_oxide_conc * (Vo * Mo / (Va * Ma + Vo * Mo))

As in same time, its elem. conc. will be set in the matrix composition

Elem conc = initial_oxide_conc * (Va * Ma / (Va * Ma + Vo * Mo))

Vo = oxide valence
Va = elem. Valence
Mo = oxygen mass (16)
Ma = Mass of the current element
```

Exemple of matrix definition for a trace processing: (Sum = 1)

7 0
11 0.0226 NL Na
..
14 0.2491 NL Si
19 0.0139 NL K
20 0.0491 NL Ca
22 0.0063 NL Ti
26 0.0680 NL Fe
…
….
8 0.4605 NL O

This TrauPIXE feature provides a more realistic matrix definition during the trace processing, improving the matrix absorption calculation and obtaining a more accurate trace composition.

Structure of an ‘external-conc.csv’ file :

View with Excel :

	A	B	C	D	E	F
1	Z	11	3	6	12	4
2	Formula	Na2O	Li	B	MgO	CO2
3	Type	PIGE	PIGE	EXT	STD	STD
4	20220825_0003_st	60155	4903	500000	1452	
5	20220825_0004_st	75000	0	500000		7500
6	20220825_0006_st	170614	0	40000	10000	
7	20220825_0007_st	37408	0	500000		
8	20220825_0008_O	20791	170	200000		
9	20220825_0009_O	21837	246	200000		
10	20220825_0010_O	18580	291	200000		

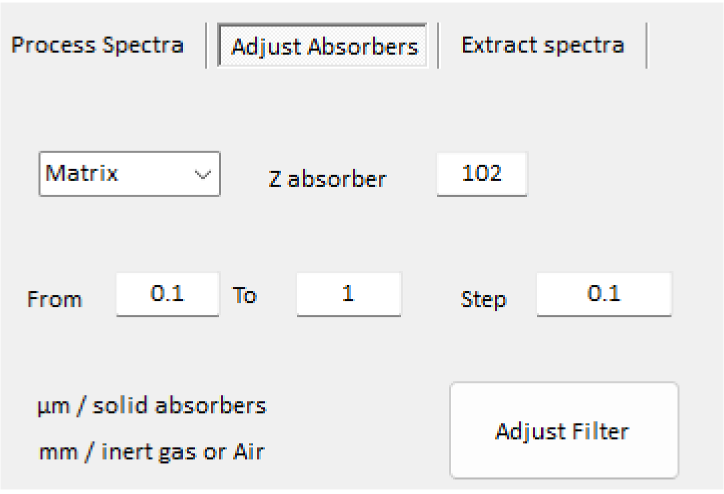
View with a Notepad :

```
Z;11;3;6;12;4;;;
Formula;Na2O;Li;B;MgO;CO2;;;
Type;PIGE;PIGE;EXT;STD;STD;;;
20220825_0003_std_VERRE_IBA;60155;4903;500000;1452;;;;
20220825_0004_std_VERRE_IBA;75000;0;500000;;7500;;;
20220825_0006_std_VERRE_IBA;170614;0;40000;10000;;
20220825_0007_std_VERRE_IBA;37408;0;500000;;;;;
20220825_0008_OA1300_VERRE_IBA;20791;170;200000;;;;;
20220825_0009_OA1300_VERRE_IBA;21837;246;200000;;;;;
20220825_0010_OA1300_VERRE_IBA;18580;291;200000;;;;;
```

XI. Automatically process thickness absorber

A new page named “Adjust Absorbers” is available.(under development May 2024)
Major development have been done since V25.08. Any filter Matrix or other traces can be adjusted.
As its name suggested, you can define one absorber present into the selected parameter file (Matrix or Trace), set the range of thickness and the step size (µm for solid and mm for gas or Air).

TrauPIXE will run simultaneously all different thickness and will provide result in a excel file



example of thickness adjustment;
Filter-Thickness-adjustment_PROJECTNAME_Filter_Matrix_Z-var-102.xlsx

	Thickness	Na2O	MgO	Al2O3	SiO2	P2O5	SO3
20230328_0002_Std_SRV_IBA	0.1	28800	44050	164600	522700	2660	27520
20230328_0002_Std_SRV_IBA	0.2	29640	44640	165400	522800	2660	27360
20230328_0002_Std_SRV_IBA	0.3	30490	45240	166200	522900	2650	27200
20230328_0002_Std_SRV_IBA	0.4	31380	45840	167000	522900	2650	27050
20230328_0002_Std_SRV_IBA	0.5	32280	46440	167800	523000	2640	26890
20230328_0002_Std_SRV_IBA	0.6	33200	47050	168500	523000	2640	26720
20230328_0002_Std_SRV_IBA	0.7	34150	47670	169300	523000	2630	26560
20230328_0002_Std_SRV_IBA	0.8	35130	48290	170100	522900	2620	26400
20230328_0002_Std_SRV_IBA	0.9	36120	48920	170900	522900	2620	26230
20230328_0002_Std_SRV_IBA	1	37150	49560	171600	522800	2610	26070